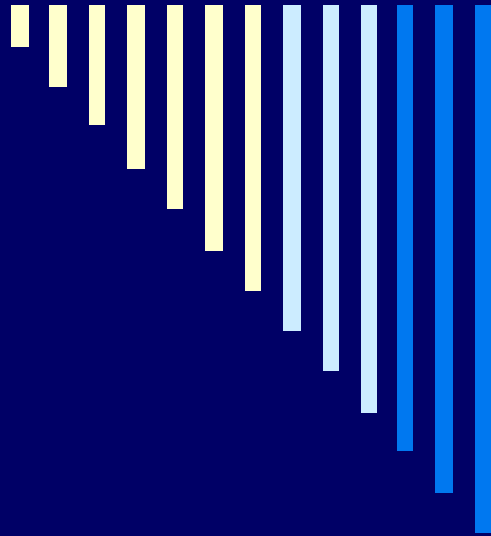
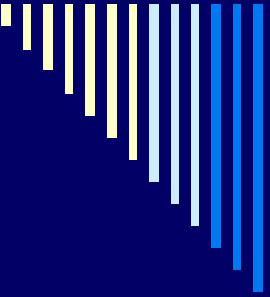


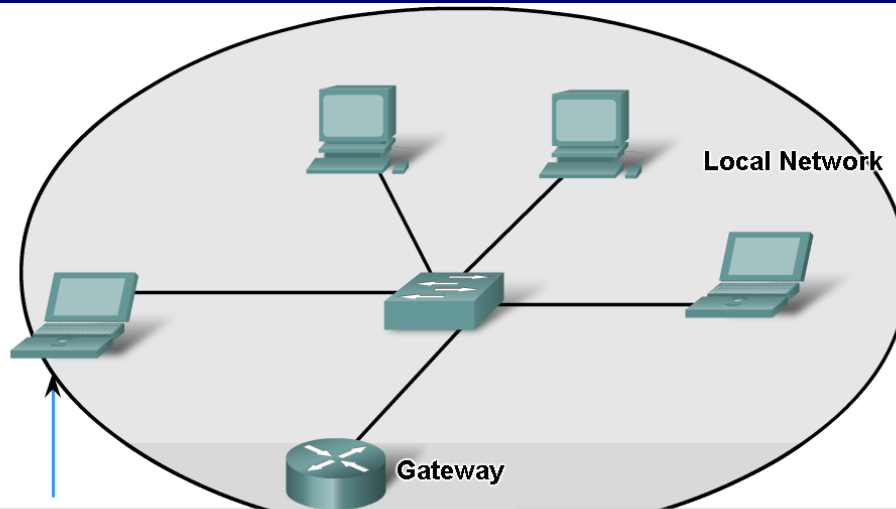


Network Layer and Routing



The Network Layer

- Layer 3 on the OSI reference model
 - The layer at which routing occurs
 - Responds to service requests from the transport layer and issues service requests to the data link layer.
-



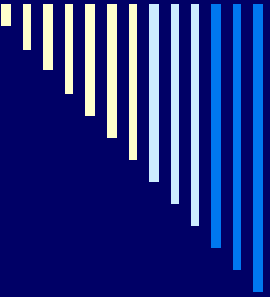
This host has the addresses for the hosts in its own network.

The address for this destination is unknown so packets are passed to the gateway router.

Hosts do not know how to deliver data to devices in a remote network - this is the role of the gateway.

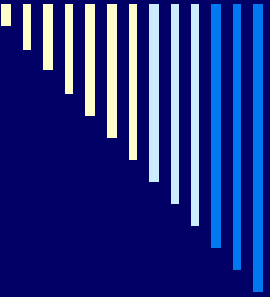
Network Layer

- ❑ Implements routing of frames (packets) through the network.
 - ❑ Defines the most optimum path the packet should take from the source to the destination
 - ❑ Defines logical addressing so that any endpoint can be identified.
 - ❑ Handles congestion in the network.
 - ❑ Facilitates interconnection between heterogeneous networks (Internetworking).
 - ❑ The network layer also defines how to fragment a packet into smaller packets to accommodate different media.
-



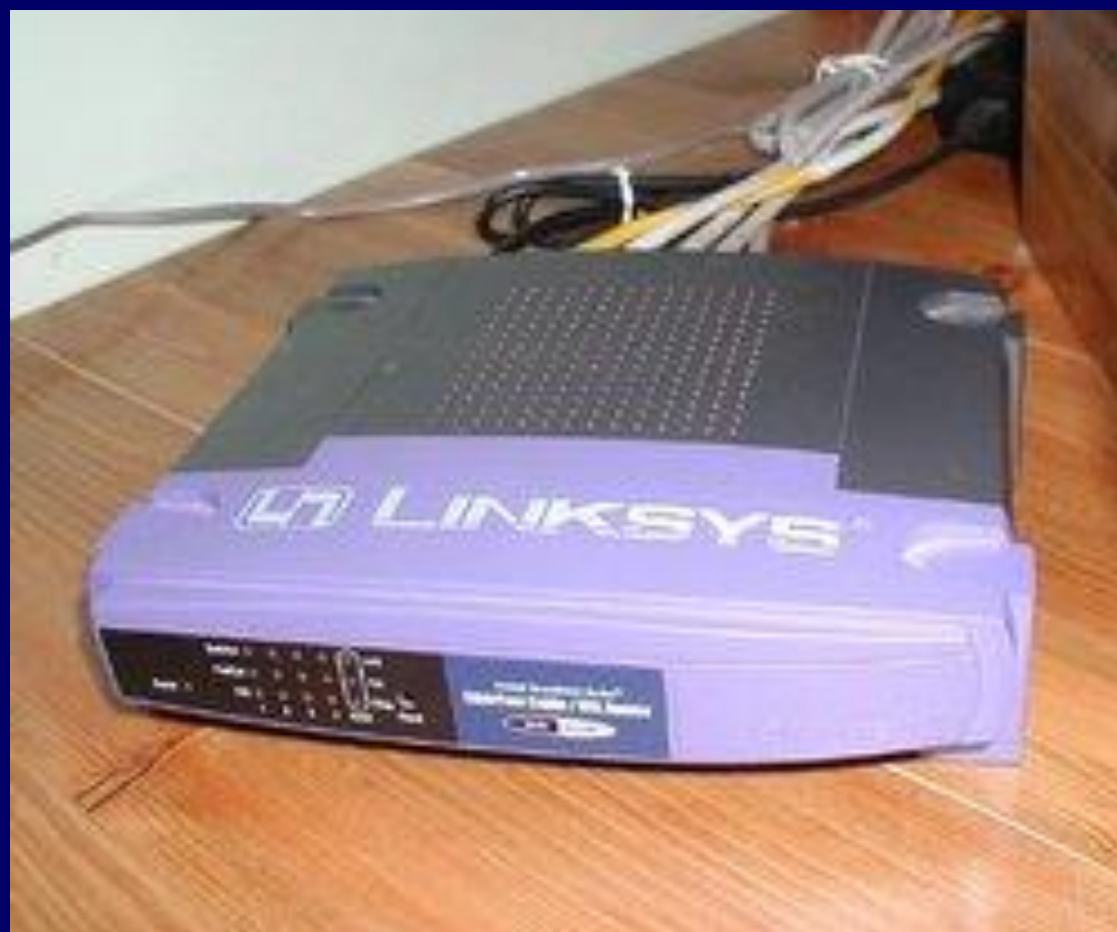
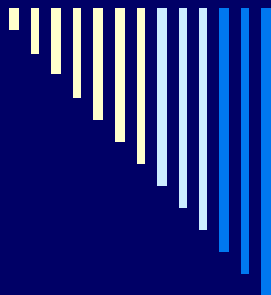
Network (Layer 3)

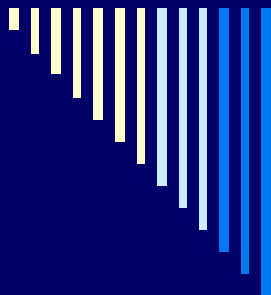
- Network layer has the following responsibilities
 - Software/logical addressing
 - Defines how data is packaged (Packets)
 - Routes data and provides connectivity
 - Best path selection
-



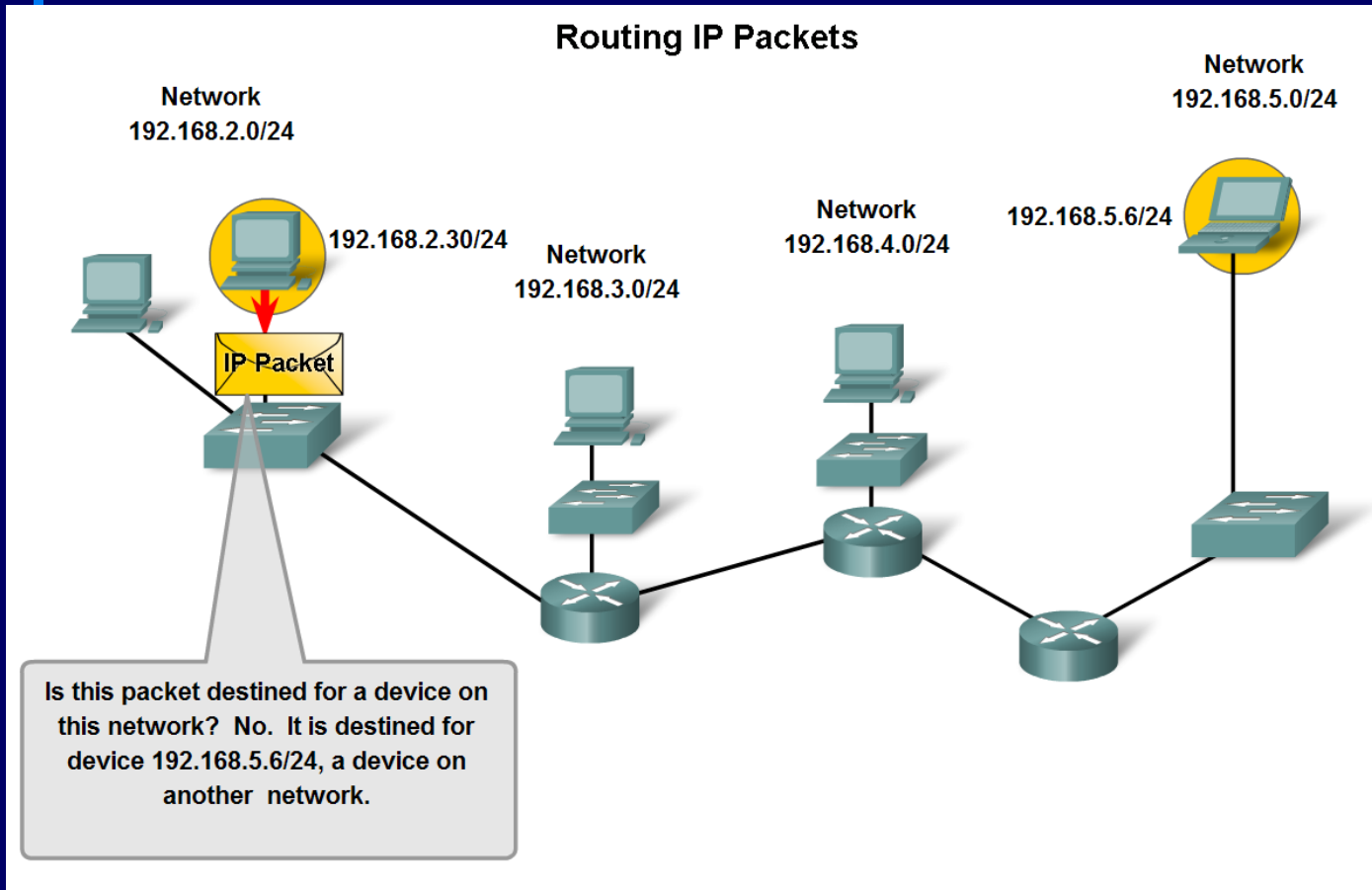
Router

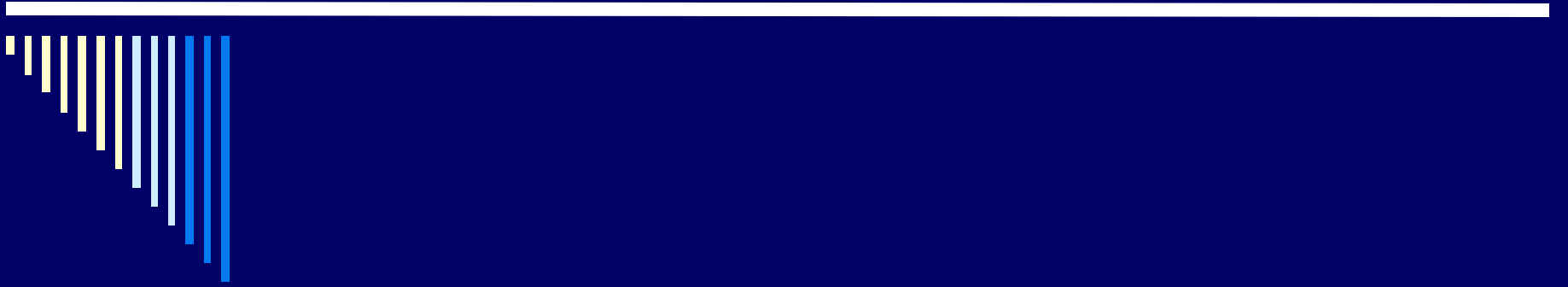
- A **router** consists of a computer networking device that determines the next network point to which a data packet has to be forwarded on its way to its destination

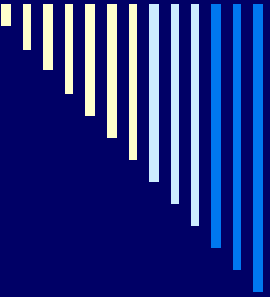




Trace the steps of an IP packet as it traverses unchanged via routers from sub network to sub-network

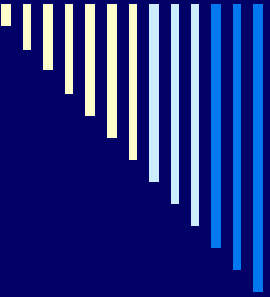






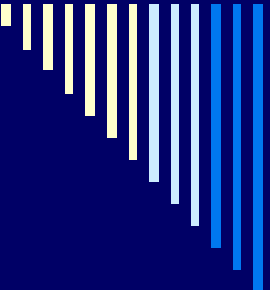
Routers

- A router is connected to at least two networks.
 - A router creates and/or maintains a table, called a "routing table" that stores the best routes to network destinations.
-



Routers

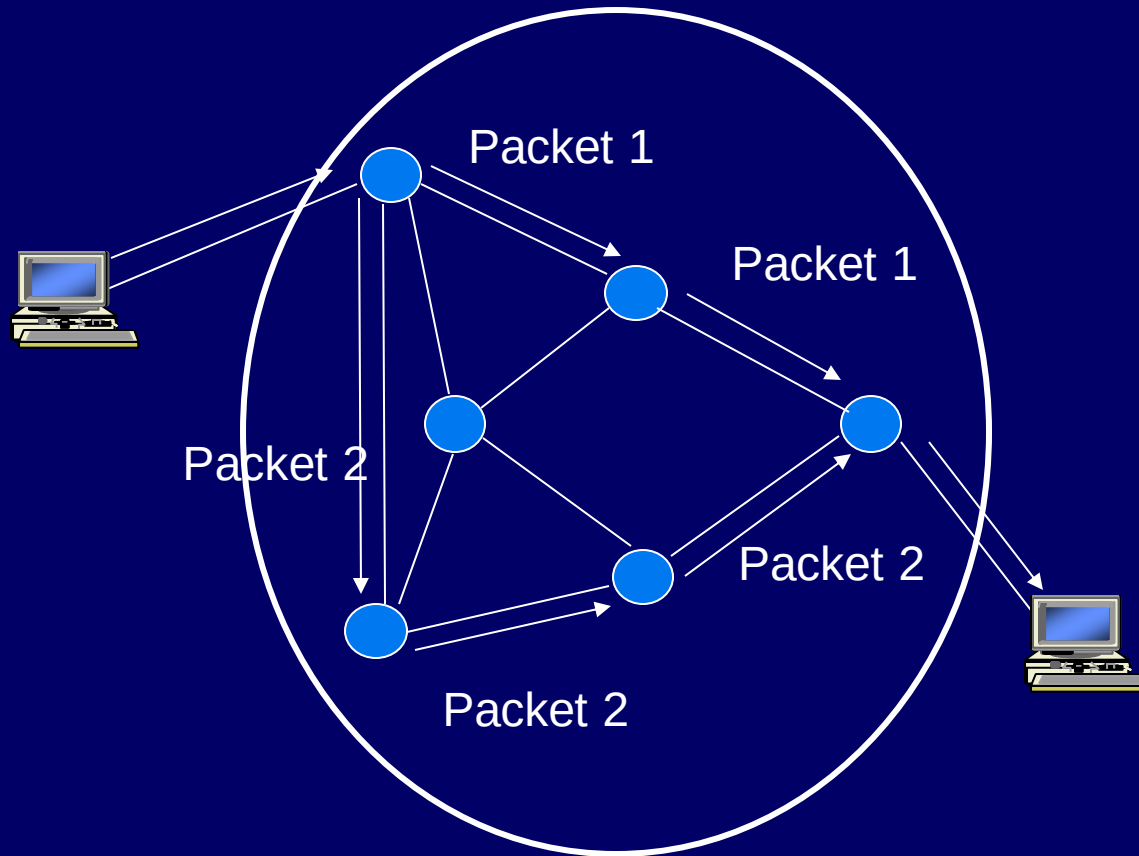
- Routing is most commonly associated with the Internet Protocol, although other less-popular routed protocols continue in use.

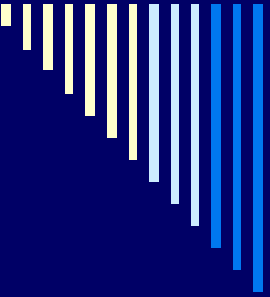


Packet Switching

- ❑ Refers to protocols in which messages are divided into packets before they are sent.
 - ❑ Each packet is then transmitted individually and can even follow different routes to its destination.
 - ❑ Once all the packets forming a message arrive at the destination, they are recompiled into the original message.
-

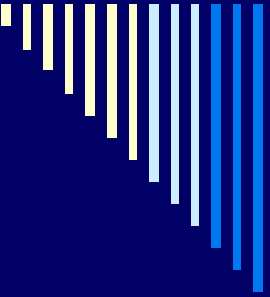
Datagram Packet Switching





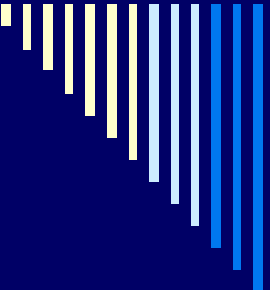
Packet switching

- *http://www.pbs.org/opb/nerds2.0.1/geek_glossary/packet_switching_flash.html*



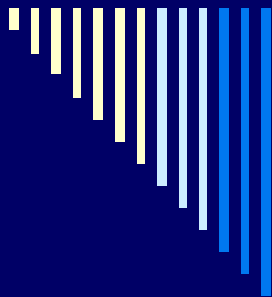
Virtual circuits

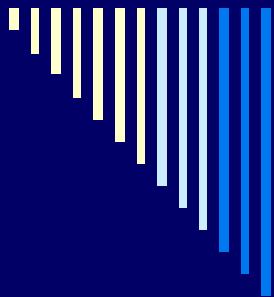
- A connection between two devices that acts as though it's a direct connection even though it may physically be circuitous.
 - The term is used most frequently to describe connections between two hosts in a packet-switching network
-

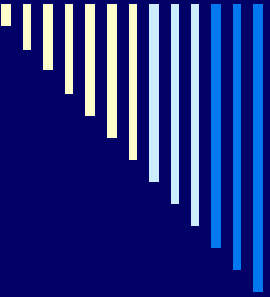


Switch

- A device that filters and forwards packets between LAN segments.
- Switches operate at the data link layer (layer 2) and sometimes the network layer (layer 3) of the OSI model and therefore support any packet protocol.

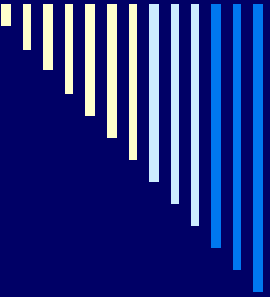






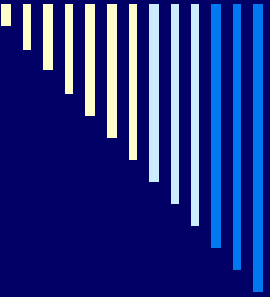
IPv4

- IPv4 is version 4 of IP. It was the first version of the Internet Protocol to be widely deployed, and forms the basis for the current (as of 2004) Internet.
- As the number of addresses available is consumed, an IPv4 address shortage appears to be inevitable in the long run.



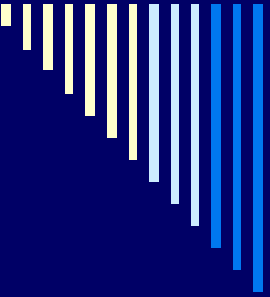
IPv6

- **IPv6** is version 6 of the Internet Protocol. IPv6 is intended to replace the previous standard, IPv4



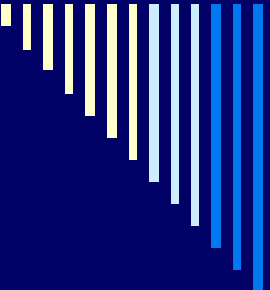
IPv6

- IPv6 is the second version of the Internet Protocol to be widely deployed, and is expected to form the basis for future expansion of the Internet.
 - The compelling reason behind the formation of IPv6 was lack of address space, especially in the heavily populated countries of Asia such as India and China.
-



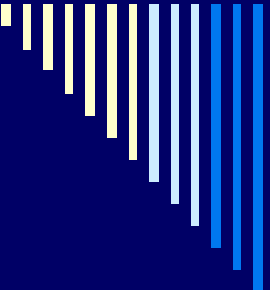
IPv6

- IPv6 addresses are 128 bits long normally written as eight groups of 4 hexadecimal digits each.
 - For example,
- 3ffe:6a88:85a3:08d3:1319:8a2e:0370:7344
- 2001:0db8:85a3:08d3:1319:8a2e:0370:7334



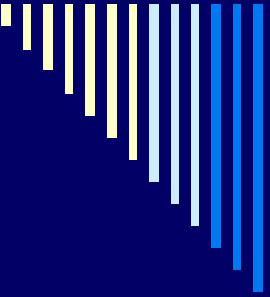
Routable and Nonroutable Protocols within the Network Layer

- ❑ An important difference between protocols is their ability to be routed.
- ❑ A protocol that is *routable* can have packets transferred across a router.
- ❑ Routable protocols require additional information be included in the packet header for routing purposes; for example, a time to live field (TTL)



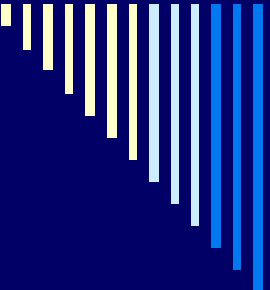
TCP/IP

- ❑ *TCP/IP (Transmission Control Protocol/Internet Protocol)* is the most common protocol used today.
 - ❑ TCP/IP, a routable protocol, is very robust and is commonly associated with UNIX systems.
 - ❑ TCP/IP was originally designed in the 1970s to be used by the Defense Advanced Research Projects Agency (DARPA) and the Department of Defense (DOD) to connect systems across the country.
-



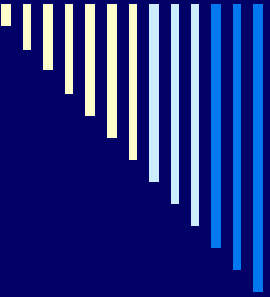
TCP/IP

- ❑ Only IP of the TCP/IP protocol resides in the Network Layer.
 - ❑ The TCP portion of the protocol is part of the Transport Layer
 - ❑ TCP is connection oriented
 - ❑ IP is connectionless
-



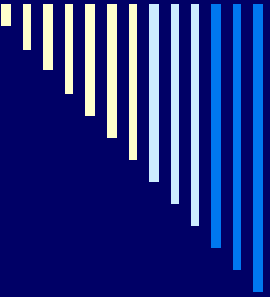
IPX/SPX

- ❑ *IPX/SPX (Internetwork Packet Exchange/Sequenced Packet Exchange)* is the protocol most commonly used with Novell NetWare.
- ❑ IPX/SPX, a routable protocol, is a very fast and highly established protocol, but it is not used on the Internet.
- ❑ The protocol IPX/SPX/NWLink was written by Microsoft and is fully compatible with the Novell protocol.



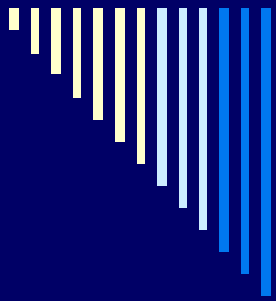
IPX/SPX

- Only IPX of the IPX/SPX protocol resides in the Network Layer.
 - The SPX portion of the protocol is part of the Transport Layer.
-



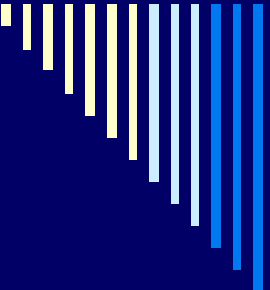
AppleTalk

- ❑ AppleTalk is the proprietary protocol developed by Apple Computer.
 - ❑ AppleTalk is rarely found in network environments where Apple Computers are not present.
 - ❑ AppleTalk is a routable protocol.
-



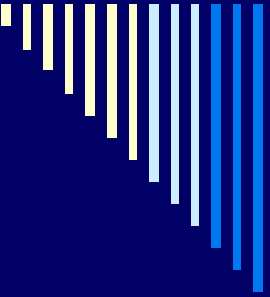
NetBEUI

- ❑ *NetBEUI (NetBIOS Extended User Interface)* is a transport protocol commonly found in smaller networks.
- ❑ NetBEUI will become less used in the future because it is not a routable protocol.
- ❑ NetBEUI is an extremely quick protocol with little overhead because of its inability to route packets.
- ❑ NetBEUI is also very easy to configure



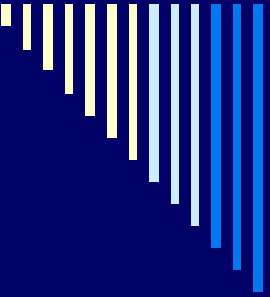
Static and Dynamic Routing

- ❑ Static routing consists of adding, maintaining, and deleting routes of the network routing devices by the network administrator.
 - ❑ Early routers had to be programmed with exactly which networks they could route between which interfaces, especially if there were many network interfaces.
 - ❑ For medium to large networks, this can be nearly impossible.
-



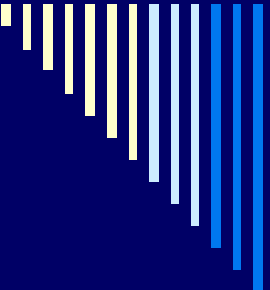
Dynamic routing

- ❑ Dynamic routing does not require the network administrator to edit complex routing tables in order to communicate with other networks or segments.
 - ❑ These routers communicate with each other using a powerful routing protocol such as Routing Information Protocol (RIP) or Open Shortest Path First (OSPF).
-



Comparing Static and Dynamic Routing

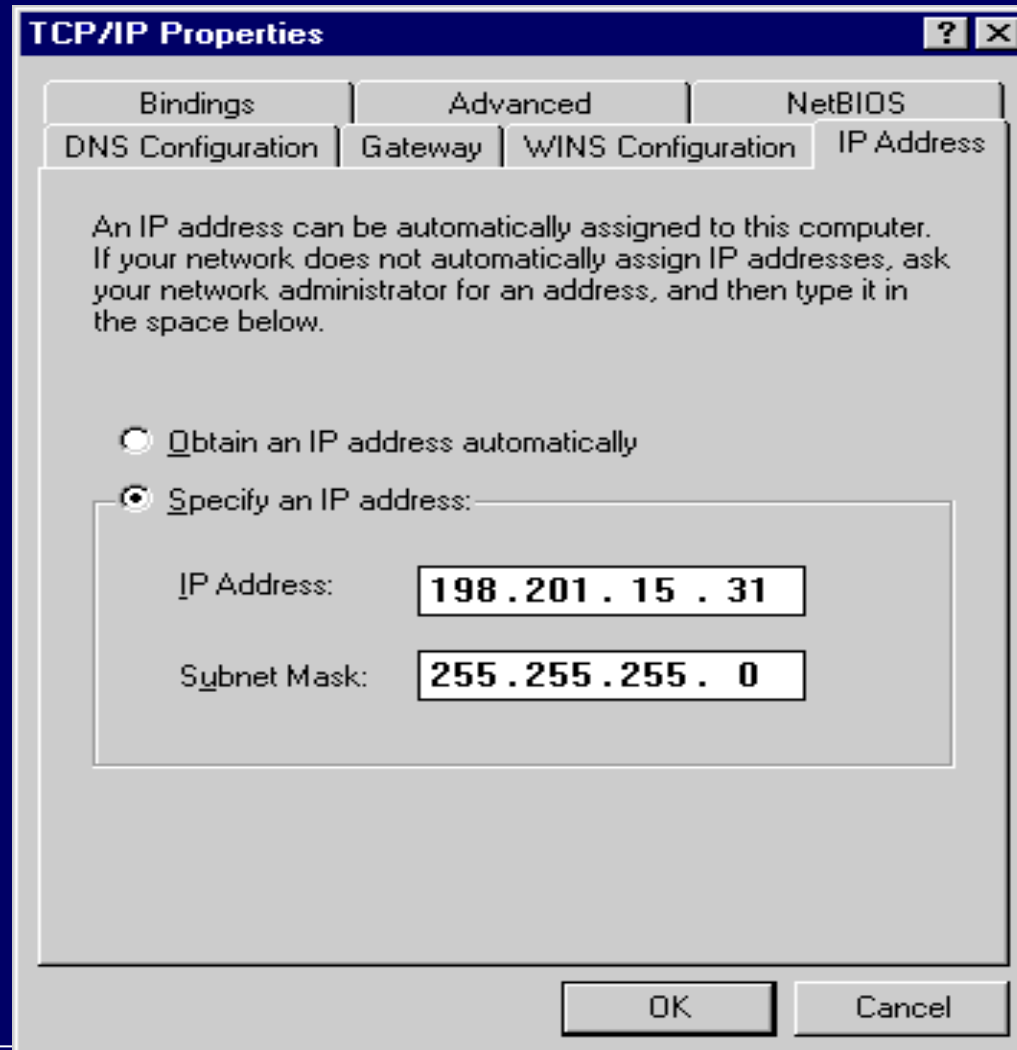
- With just two networks, the static routing setup is the more appropriate.
- If your network has several parallel networks, dynamic routing would be easier to set up.



Default Gateways

- The default gateway is specified on each computer, and sends the packet to the first router.
 - When the packet hits this first router, the router must determine if the destination computer is on the local network, or send the packet to the next router that will get the packet to its destination.
-

IP address and subnet mask



The image shows a screenshot of the 'TCP/IP Properties' dialog box from a Windows operating system. The dialog box has a title bar with a question mark and a close button. It contains several tabs: 'Bindings', 'Advanced', 'NetBIOS', 'DNS Configuration', 'Gateway', 'WINS Configuration', and 'IP Address'. The 'IP Address' tab is currently selected. Inside the dialog, there is a text box with the following text: 'An IP address can be automatically assigned to this computer. If your network does not automatically assign IP addresses, ask your network administrator for an address, and then type it in the space below.' Below this text, there are two radio buttons. The first radio button is labeled 'Obtain an IP address automatically' and is unselected. The second radio button is labeled 'Specify an IP address:' and is selected. Below the second radio button, there are two text input fields. The first field is labeled 'IP Address:' and contains the text '198 . 201 . 15 . 31'. The second field is labeled 'Subnet Mask:' and contains the text '255 . 255 . 255 . 0'. At the bottom of the dialog box, there are two buttons: 'OK' and 'Cancel'.

TCP/IP Properties

Bindings | Advanced | NetBIOS
DNS Configuration | Gateway | WINS Configuration | **IP Address**

An IP address can be automatically assigned to this computer. If your network does not automatically assign IP addresses, ask your network administrator for an address, and then type it in the space below.

☐ Obtain an IP address automatically

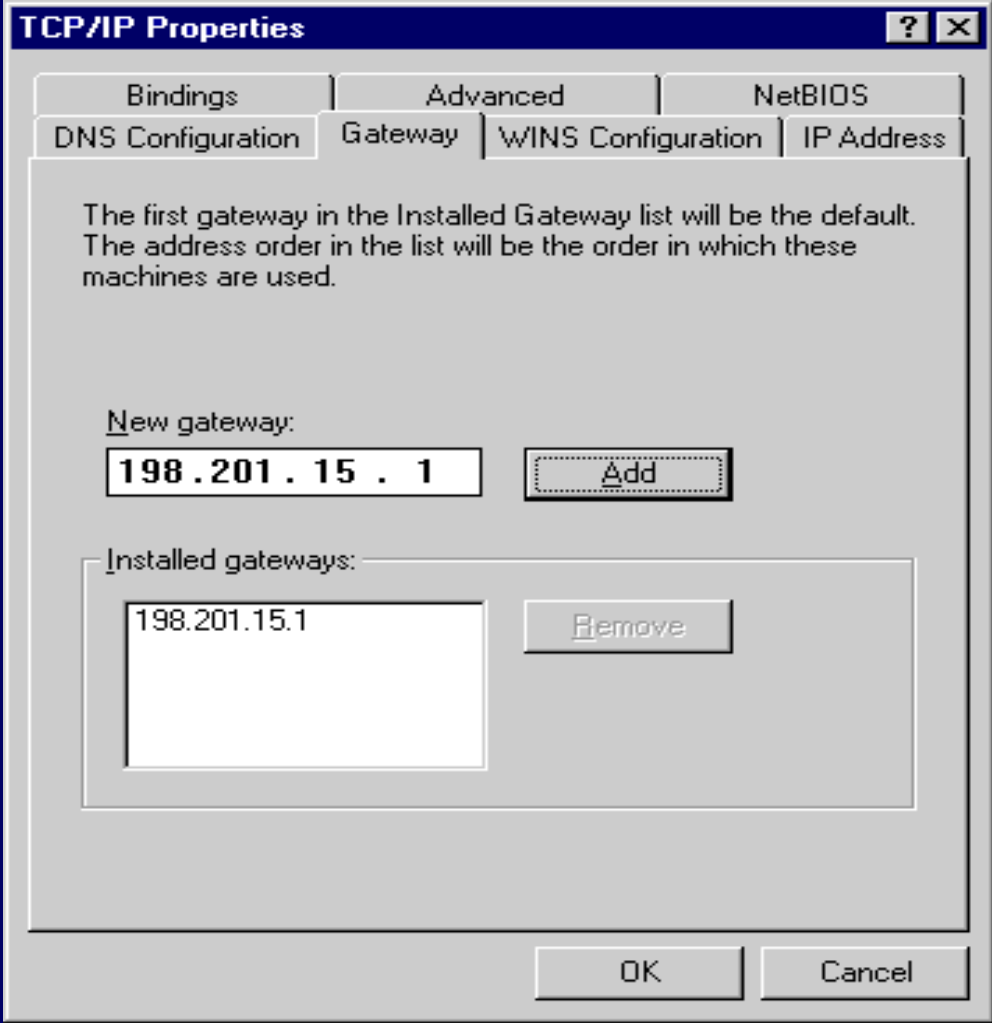
☒ Specify an IP address:

IP Address: **198 . 201 . 15 . 31**

Subnet Mask: **255 . 255 . 255 . 0**

OK Cancel

Default gateway



The screenshot shows the 'TCP/IP Properties' dialog box with the 'Gateway' tab selected. The dialog has a title bar with a question mark and a close button. Below the title bar are three tabs: 'Bindings', 'Advanced', and 'NetBIOS'. Under 'Advanced', there are four sub-tabs: 'DNS Configuration', 'Gateway' (selected), 'WINS Configuration', and 'IP Address'. The main area contains instructional text about gateway selection, a 'New gateway' section with an input field containing '198.201.15.1' and an 'Add' button, and an 'Installed gateways' section with a list box containing '198.201.15.1' and a 'Remove' button. At the bottom are 'OK' and 'Cancel' buttons.

TCP/IP Properties [?] [X]

Bindings | Advanced | NetBIOS

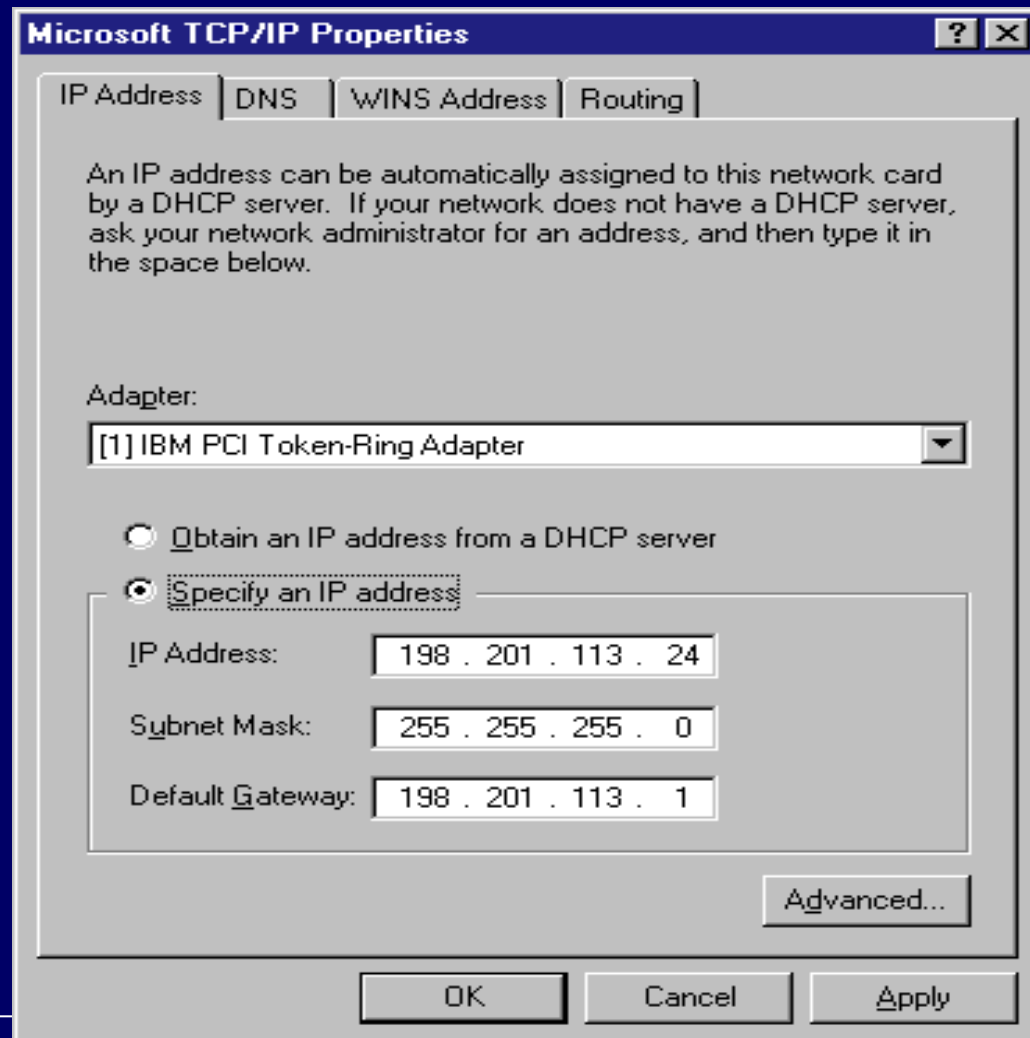
DNS Configuration | **Gateway** | WINS Configuration | IP Address

The first gateway in the Installed Gateway list will be the default.
The address order in the list will be the order in which these machines are used.

New gateway:

Installed gateways:

Configuring IP address, subnet mask and default gateway



The screenshot shows the 'Microsoft TCP/IP Properties' dialog box with the 'IP Address' tab selected. The dialog box has a title bar with a question mark and a close button. Below the title bar are four tabs: 'IP Address', 'DNS', 'WINS Address', and 'Routing'. The 'IP Address' tab is active and contains the following elements:

- A text box with the instruction: "An IP address can be automatically assigned to this network card by a DHCP server. If your network does not have a DHCP server, ask your network administrator for an address, and then type it in the space below."
- A label 'Adapter:' followed by a dropdown menu showing '[1] IBM PCI Token-Ring Adapter'.
- Two radio buttons for IP assignment:
 - ☐ Obtain an IP address from a DHCP server
 - ☒ Specify an IP address
- A group box containing three input fields:
 - 'IP Address:' with the value '198 . 201 . 113 . 24'
 - 'Subnet Mask:' with the value '255 . 255 . 255 . 0'
 - 'Default Gateway:' with the value '198 . 201 . 113 . 1'
- An 'Advanced...' button at the bottom right of the group box.

At the bottom of the dialog box are three buttons: 'OK', 'Cancel', and 'Apply'.